

NICHOLAS W. LUNDGREN

Process Engineer | Semiconductor Materials | Data-Driven Characterization
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PROFESSIONAL SUMMARY

PhD candidate in physical chemistry with 5+ years of research in semiconductor thermal transport, materials characterization, and computational modeling. Experienced in designing experiments, analyzing large datasets, troubleshooting complex systems, and translating findings into clear technical documentation. Proven ability to mentor junior researchers, communicate across disciplines, and implement new analytical tools. Seeking to apply deep materials science knowledge and process optimization skills in a process engineering role.

CORE COMPETENCIES

Data Analysis and Visualization (Python)	Machine Learning and Scientific Computing	Cross-functional Communication
HPC Workflows and Linux Systems	Raman and Brillouin Spectroscopy	Mentorship and Team Leadership
Molecular Dynamics and Lattice Dynamics	Technical Documentation	Python, Fortran, C++, Git, LaTeX

EXPERIENCE

Graduate Researcher , Department of Chemistry, UC Davis	Sep 2020 to Present
- Designed and executed thermal transport experiments on semiconductor and amorphous materials using molecular dynamics, lattice dynamics, and machine learning potentials, analyzing results to inform materials design. - Performed characterization of low-symmetry and disordered material systems, interpreting film and bulk properties through computational methods. - Diagnosed and resolved complex simulation failures across HPC platforms through systematic root cause analysis, including debugging software builds, compiler issues, and architecture-specific bottlenecks. - Measured and interpreted thermal conductivity, vibrational spectra, and phonon transport properties in thin-film silicon, amorphous carbon, and 2D allotropes (silicene). - Authored peer-reviewed publications, conference presentations, and internal reports documenting novel modeling approaches and experimental findings. - Co-developed the open-source kALDo thermal transport code, implementing new features, automating documentation pipelines, and evaluating third-party tools for integration. - Trained machine learning interatomic potentials to enable large-scale simulations.	
Teaching Assistant , Department of Chemistry, UC Davis	2020 to 2026
- Communicated complex technical material to diverse audiences across general chemistry, thermodynamics, spectroscopy, quantum chemistry, kinetics, and organic chemistry courses. - Mentored an undergraduate researcher through experimental design and data interpretation, serving as a technical resource for junior colleagues. - Nominated for Best Teaching Assistant 2025 for excellence in communication, and timely execution of duties	

EDUCATION

PhD in Chemistry (Physical Chemistry) , University of California, Davis	Expected June 2026
Focus: Thermal transport in semiconductor and disordered materials; scientific software development; ML for atomistic modeling. GPA: 3.75	
BS in Chemical Physics, Highest Honors , University of California, Davis	June 2020
Thesis: The Effects of Germanium Concentration on the Thermal Transport of Amorphous Silicon. GPA: 3.61	

SELECTED PUBLICATIONS

- Layer-Stacking and Strain Effects on Heat Transport in Silicene, Submitted 03/2026 (PRB) - kALDo 2.0: Thermal Transport from Machine Learning Potentials, Computer Physics Communications (2026, accepted) - Mode Localization and Suppressed Heat Transport, Physical Review B (2021) - Efficient Anharmonic Lattice Dynamic Calculations, Journal of Applied Physics (2020)	
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TECHNICAL TOOLS

Software: LAMMPS, GPUMD, DLPOLY, ASE, kALDo, TensorFlow, Kokkos, VMD, Inkscape, Git, LaTeX
Languages: Python, Fortran, JavaScript, HTML/CSS | Spoken: English (native), Portuguese (fluent)